

RESEARCH ABSTRACT

MACHINE LEARNING-BASED INTERACTIVE MOBILE APPLICATION FOR EARLY RISK DETECTION AND MIDWIFERY SUPPORT

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This dissertation presents the design, development, and evaluation of a machine learning-based interactive mobile application for early maternal risk detection and midwifery support in the Sri Lankan antenatal care context. The study responds to limitations in paper-based antenatal workflows, where midwives must manually interpret vital parameters under time pressure and limited connectivity.

The proposed system uses supervised machine learning models, including Random Forest, Support Vector Machine, XGBoost, and CNN/LSTM, to classify maternal risk using parameters such as blood pressure, blood sugar, body temperature, fetal heart rate, and hemoglobin. The selected deployment approach combines real-time on-device inference with interactive dashboards, SHAP/LIME-based explainability, trilingual interface support, and voice and haptic alerts.

The methodology included requirement validation with practicing midwives, data preprocessing, model evaluation, Flutter-based mobile application design, and usability testing. SMOTE was applied only to the training partition to avoid test-set data leakage. The deployed ensemble model achieved 90.1% accuracy with an average inference latency of 1.4 seconds on mid-range Android hardware. Usability evaluation with 18 midwifery and nursing students produced a mean SUS score of 78.5, while clinical simulation trials showed improved decision speed and fewer high-risk identification errors compared with the paper-based workflow.

The findings suggest that mobile machine learning decision support can improve the speed, confidence, and consistency of antenatal risk assessment when used as an assistive tool rather than a replacement for professional clinical judgement.

Keywords: *mHealth; machine learning; maternal risk prediction; midwifery; Sri Lanka*